

Q2 (5 points)

1 page submitted

Find the domain of the function

$$f(x) = \frac{\sqrt{\ln(4-x)}}{\cos^{-1}(2x-1)}$$

Write the answer

a) in the set notation

b) in the interval notation

Q3 (10 points)

1 page submitted

Find all roots of the given trigonometric equation

$$\cos 2x = -\sin x$$

Q4 (3 points)

1 page submitted

Determine whether the given statement is true or false. Justify your answer

a) If a number is divisible by 6, then it is divisible by 3.

Q5 (3 points)

1 page submitted

Determine whether the given statement is true or false. Justify your answer

b) For all real numbers x and y , $\frac{x}{y} = 0$ if and only if $x = 0$.

Q6 (3 points)

1 page submitted

Determine whether the given statement is true or false. Justify your answer

c) For all real numbers y there is a real number x , such that $y = 2x + 4$

Q7 (3 points)

1 page submitted

Determine whether the given statement is true or false. Justify your answer

d) For all real numbers $x > 0$ and $y > 0$, if $x > y$, then $\frac{1}{x} < \frac{1}{y}$

Q8 (8 points)

1 page submitted

Write the converse and contrapositive of the statement

"If x is divisible by 12, then x is divisible by 3"

Determine the truth of the statement, its converse, and its contrapositive.

Q9 (10 points)

1 page submitted

Prove that

$$\log_b xy = \log_b x + \log_b y$$

Q10 (10 points)

1 page submitted

Prove that $f(x) = 2x^2 + x - 1$ is not one-to-one function.

Q11 (10 points)

2 pages submitted

Write the following limits as a formal statement involving δ , ϵ or M . Sketch the graph that illustrates the roles of these constants.

a) $\lim_{x \rightarrow 3} (x^2 - 4) = 5$

b) $\lim_{x \rightarrow 2^+} \frac{4}{x-2} = \infty$

Q12 (10 points)

1 page submitted

Given $\lim_{x \rightarrow 2} (2x + 1) = 5$. Approximate the value of δ that works for $\epsilon = 0.1$

Q13 (10 points)

1 page submitted

Show that $\lim_{x \rightarrow 2.5} (x^2 - 1) \neq 2$.

Q14 (10 points)

1 page submitted

Given $\lim_{x \rightarrow \infty} \frac{x+1}{x} = 1$ Use the graph and algebra to approximate the smallest N that guarantees that if $x \in (N, \infty)$ then $f(x) \in (0.75, 1.25)$.

BONUS (The mark for this question will be added to the total mark. The total mark can be 110%)

Find the composition and the domain of the composition $f \circ g$. Draw the graph of the resulting function.

$$f(x) = \begin{cases} 1 - x & x \leq 0 \\ x^2 & x > 0 \end{cases}$$

$$g(x) = \begin{cases} -x & x < 1 \\ x + 1 & x \geq 1 \end{cases}$$